Demo: Speaker

From changing current to motion and sound

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

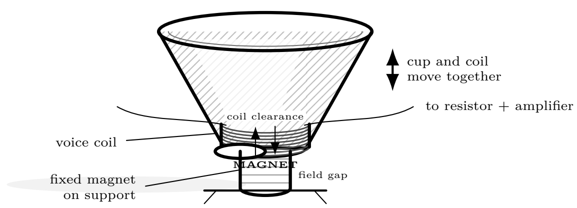
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Paper cup
- 30–32 AWG insulated wire coil
- Ceramic or secured neodymium magnet
- Audio source through approved battery amplifier
- 100 Ω series resistor

Predict first

Will steady DC make a continuous tone? Why should alternating current differ?



Investigate

1. Wind and secure a light coil to the cup base. Adult secures the magnet so it cannot jump or be ingested.
2. Connect only through the resistor and a battery-powered, current-limited amplifier at minimum level.
3. Play a low-volume tone and observe cup motion by touching a paper flag, not the moving coil.
4. Change frequency, then mass with a small removable paper clip; compare sound.

Explain from the evidence

The amplifier supplies changing current. In the permanent magnet's field, coil force reverses with current and moves the cone, which drives pressure variations in air. Steady current gives only a displacement and heating, not a sustained tone. Electrical, mechanical, and acoustic power are linked; losses become heat.

Change one thing

Use a phone spectrum display only as an observer. Compare commanded frequency with the strongest measured peak.

Evidence record

Prediction	______________________________
One change	______________________________
Observation	______________________________
Claim + because	______________________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.